

 COMPUTER VISION & HEALTHCARE AI

IDC Detection from Breast Histopathology Images

Deep Learning-Based Image Classification Using ResNet-18

PRESENTED BY

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DOMAIN

Data Science | Deep Learning | Computer Vision

Python

PyTorch

ResNet-18

Scikit-learn

Why I Made This Project

THE HEALTHCARE CONCERN

98,337

Deaths in India (2022 GLOBOCAN)

84.6%

Catastrophic Health Spend

Financial Strain: 72.4% experienced distress financing via loans or asset sales. WHO highlights global diagnosis inequality.

Dataset Note

Benchmark dataset used to demonstrate AI methodology due to availability constraints of recent public labeled sets.

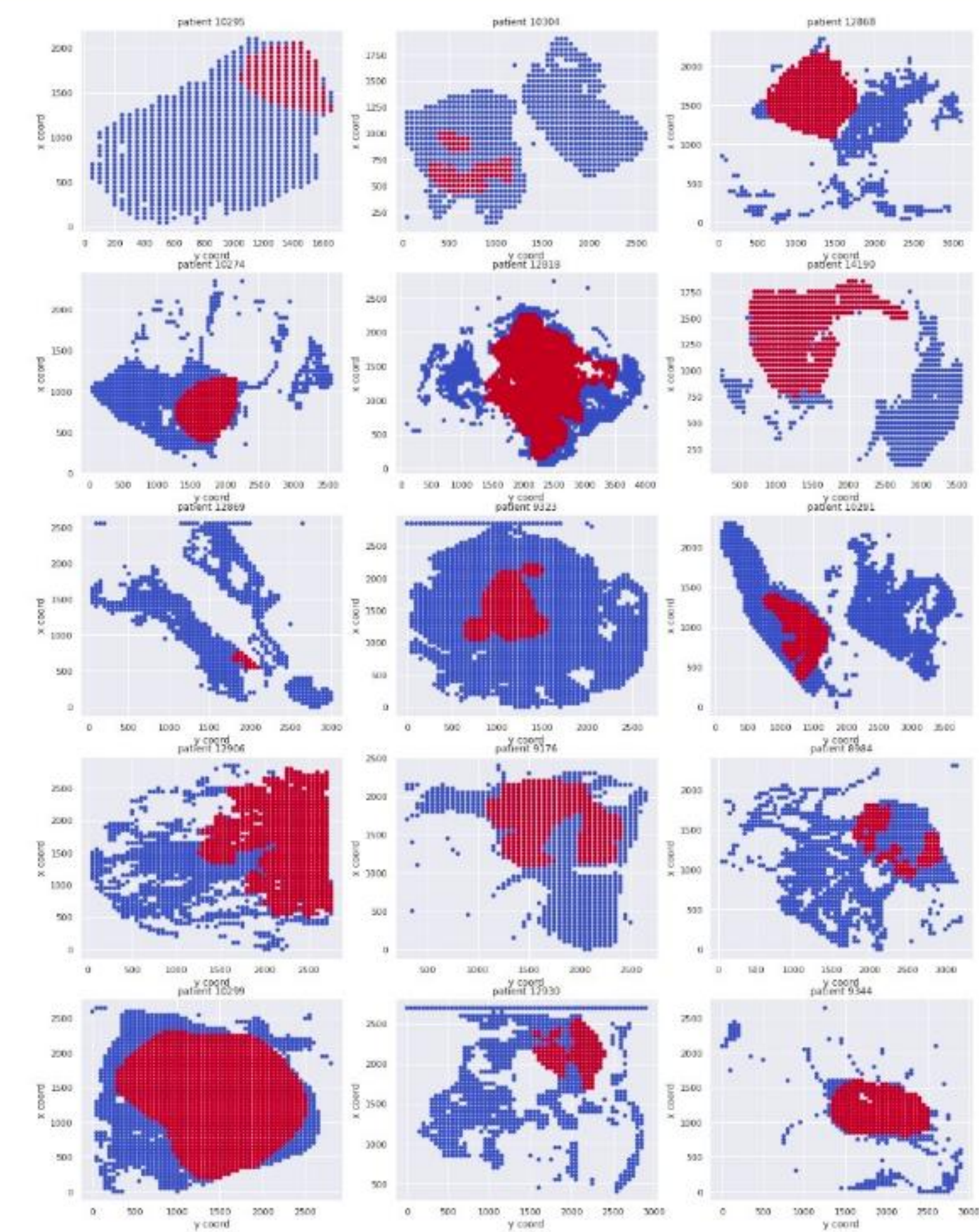
My Motivation

Exploring deep learning for accessible, automated tissue analysis to support affordable local healthcare solutions in India.

 Focuses on patch-level IDC classification; does not replace clinical confirmation.

Introduction

- **Complex Histopathology:** Microscopic tissue structures contain complex patterns that are difficult to analyze manually at scale.
- **Invasive Ductal Carcinoma (IDC):** Most common form of breast cancer, originating in the milk ducts.
- **Computer Vision Solution:** Automates tissue patch classification into binary targets (IDC vs Non-IDC).
- **End-to-End Workflow:** Covers data exploration, patch preprocessing, ResNet-18 modeling, and tissue reconstruction.



Problem Statement

PRIMARY OBJECTIVE

Develop a deep learning classification model for breast histopathology image patches:

Class 0 → Non-IDC (Healthy / Benign)

Class 1 → IDC (Malignant Invasive Ductal)

Scale & Volume

Processing ~277k micro image patches efficiently.

Class Imbalance

Significantly more Non-IDC patches than IDC patches.

Data Leakage Risk

Requires patient-level separation to evaluate generalization.

Dataset Overview & Patient Dependency

Feature	Dataset Details
Total Image Patches	~277,524 patches
Patient Count	279 patients
Patch Size	50 × 50 pixels (RGB)
Problem Type	Binary Image Classification
Classes	Non-IDC (0) vs IDC (1)

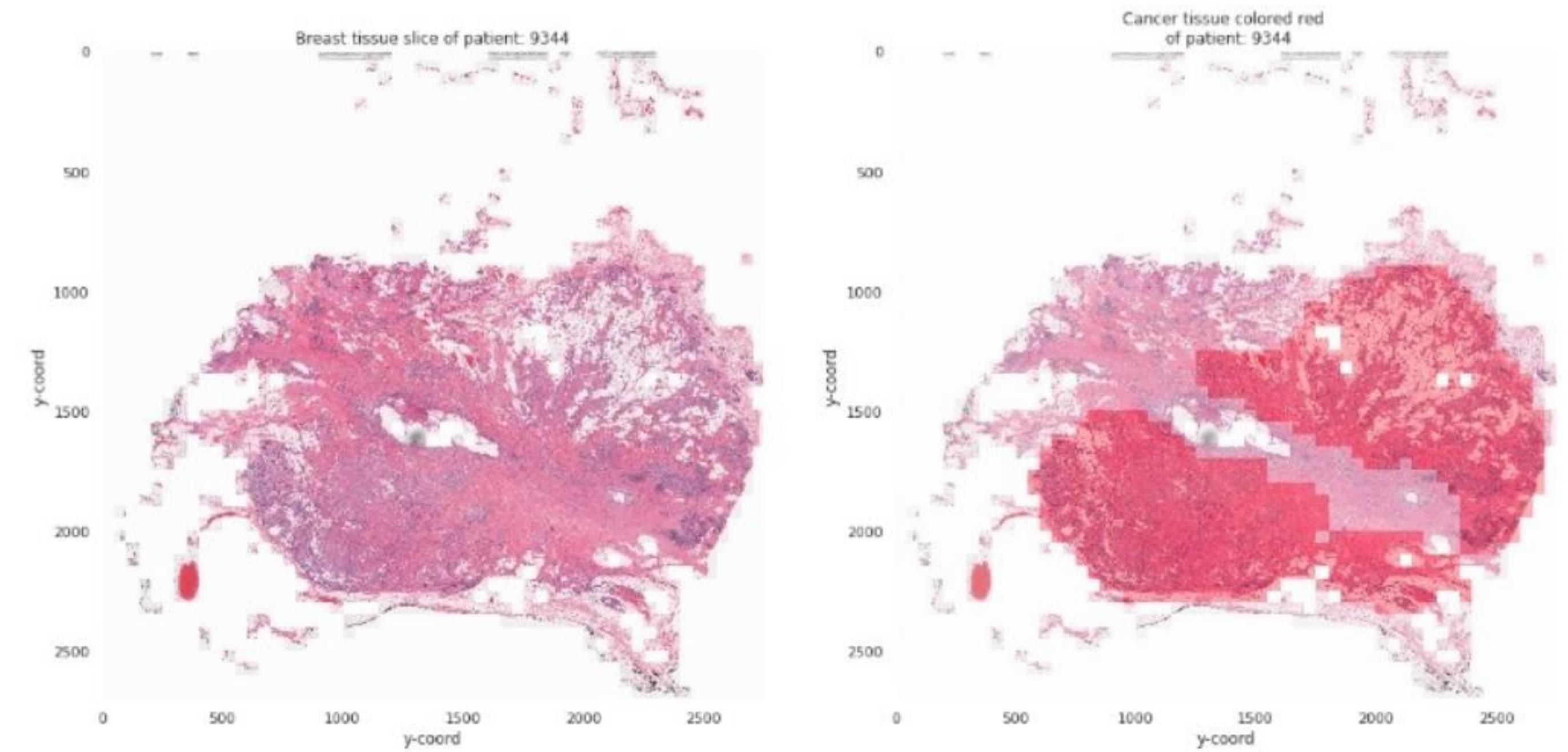
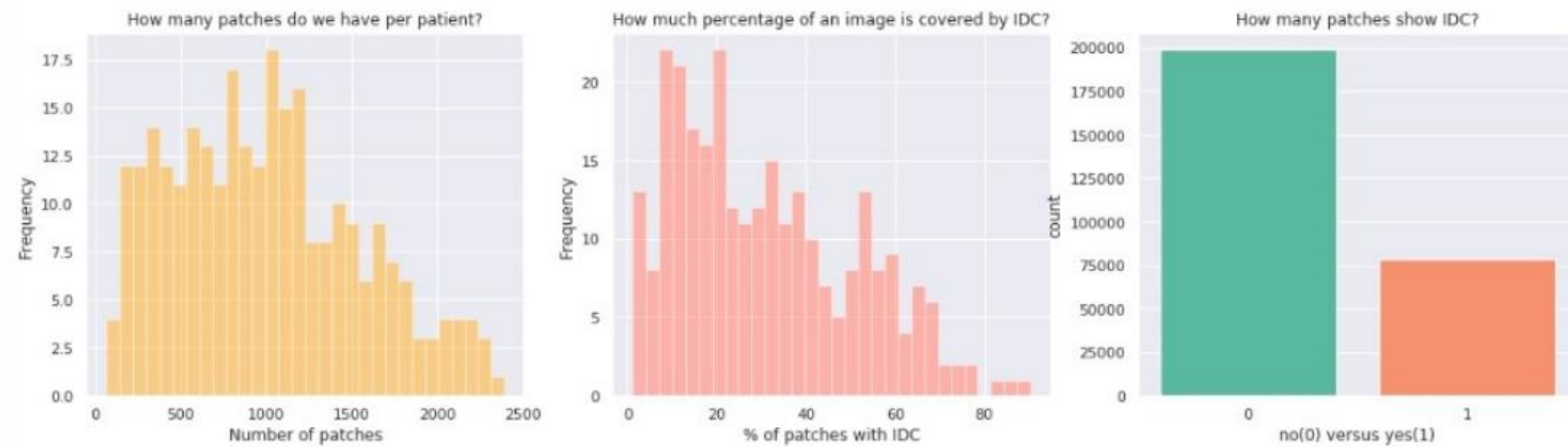
 Critical Observation

Each patient contributes hundreds of individual image patches.

Key Implication:

Patches from the same patient are correlated. Splitting at the patient level is mandatory to prevent data leakage.

Exploratory Data Analysis (EDA)



Class & Spatial Insights

Significant class skew (~71% Non-IDC vs ~29% IDC). Cancerous patches cluster in core tumor centers rather than uniform dispersion.

Key Finding

Class imbalance requires weighted loss strategies during training to prevent biased majority predictions.

Data Preprocessing & Augmentation

1. Image Standardization

Load RGB tissue patches → Resize to 50×50 px → Channel normalization.

2. Training Augmentation

Random Horizontal & Vertical Flips to increase dataset diversity and generalization.

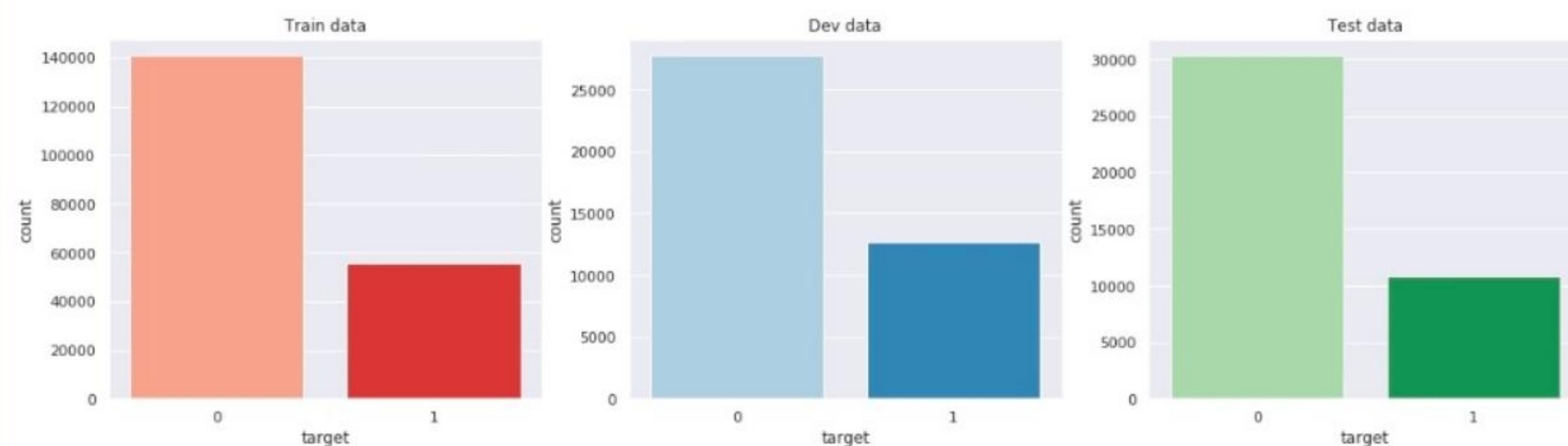
3. Spatial Coordinate Tracking

Retain (x, y) coordinates and patient IDs for downstream slide reconstruction.

dev_predictions.head() - Patch Coordinates

true	predicted	proba	x	y	patient_id
0	0	0.0034	501.0	2001.0	9135
0	0	0.1114	1251.0	501.0	9135
0	0	0.2350	351.0	351.0	9135
0	0	0.5992	351.0	1051.0	9135
0	0	0.0967	1401.0	151.0	9135

Patient-Level Data Splitting



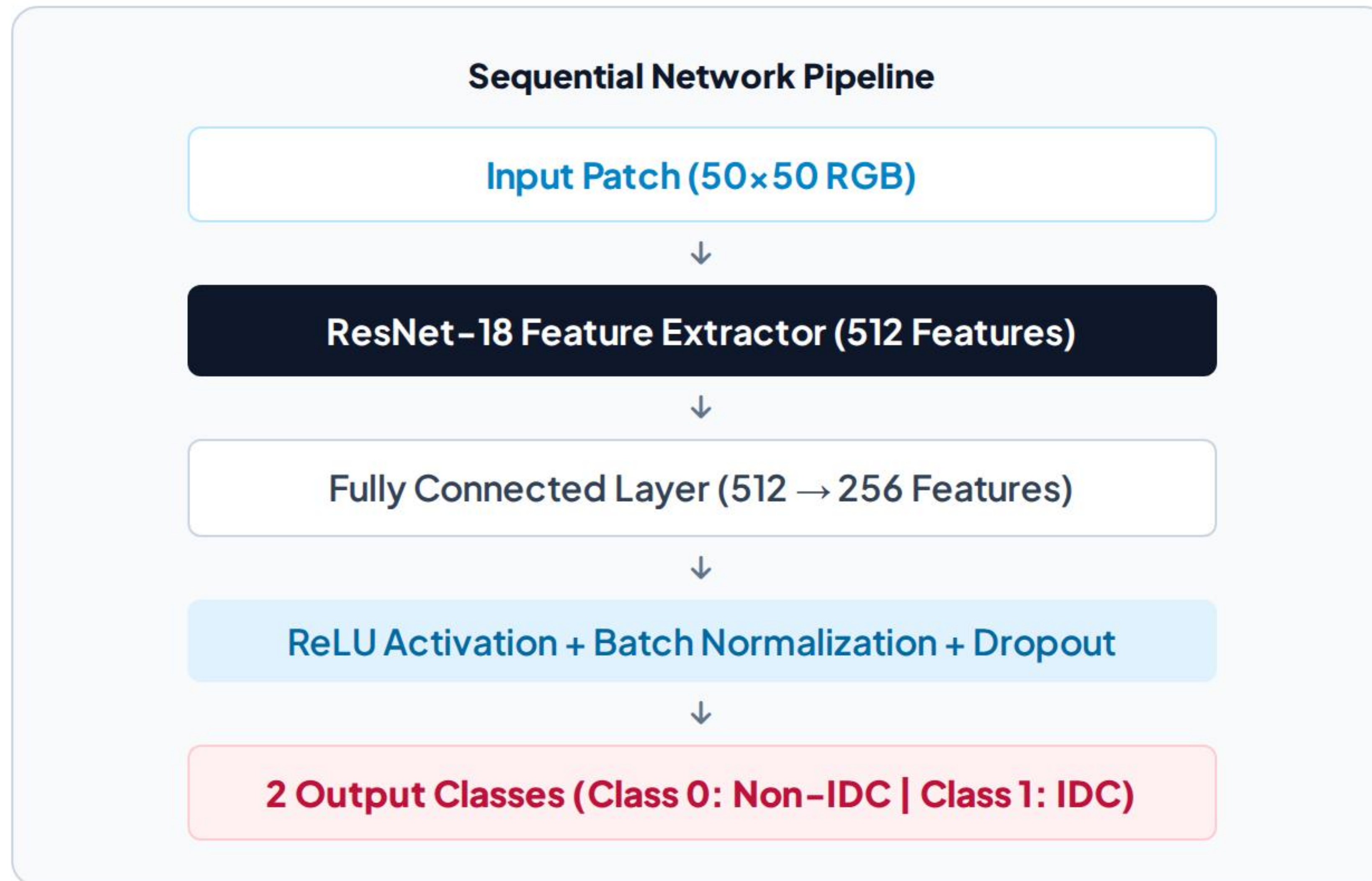
Patient-Level Split Rationale

- **Training Set (~70% Patients):** Primary model optimization.
- **Dev Set (~15% Patients):** Hyperparameter tuning & LR search.
- **Test Set (~15% Patients):** Final unseen evaluation.

Preventing Data Leakage

Patches from the same patient share visual characteristics. Splitting strictly by patient ID ensures realistic performance on unseen tissues.

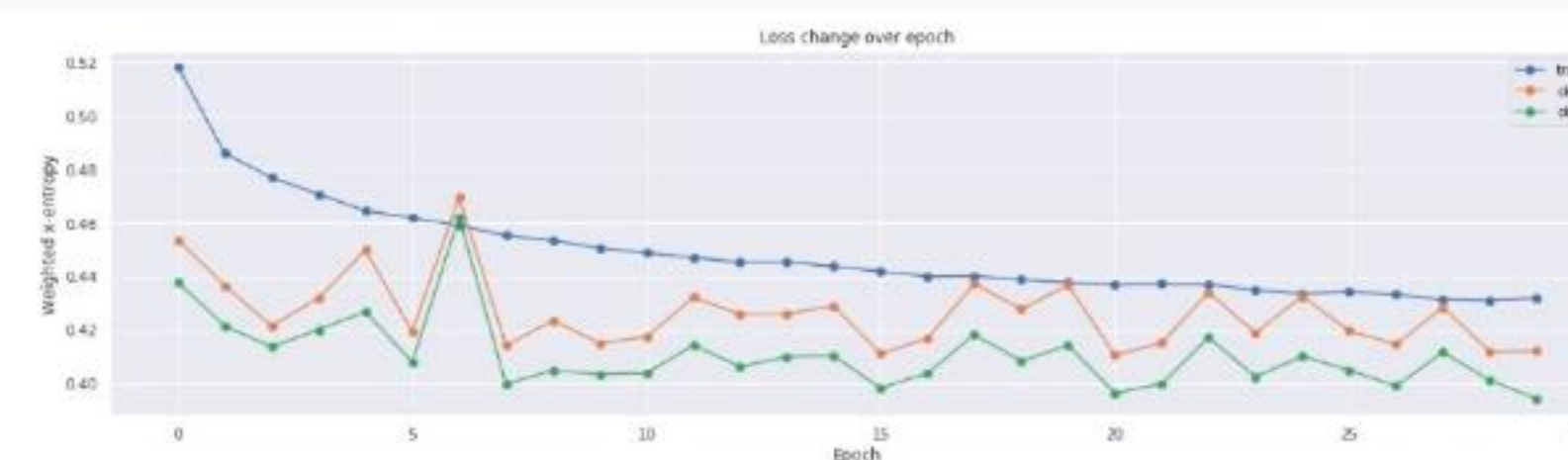
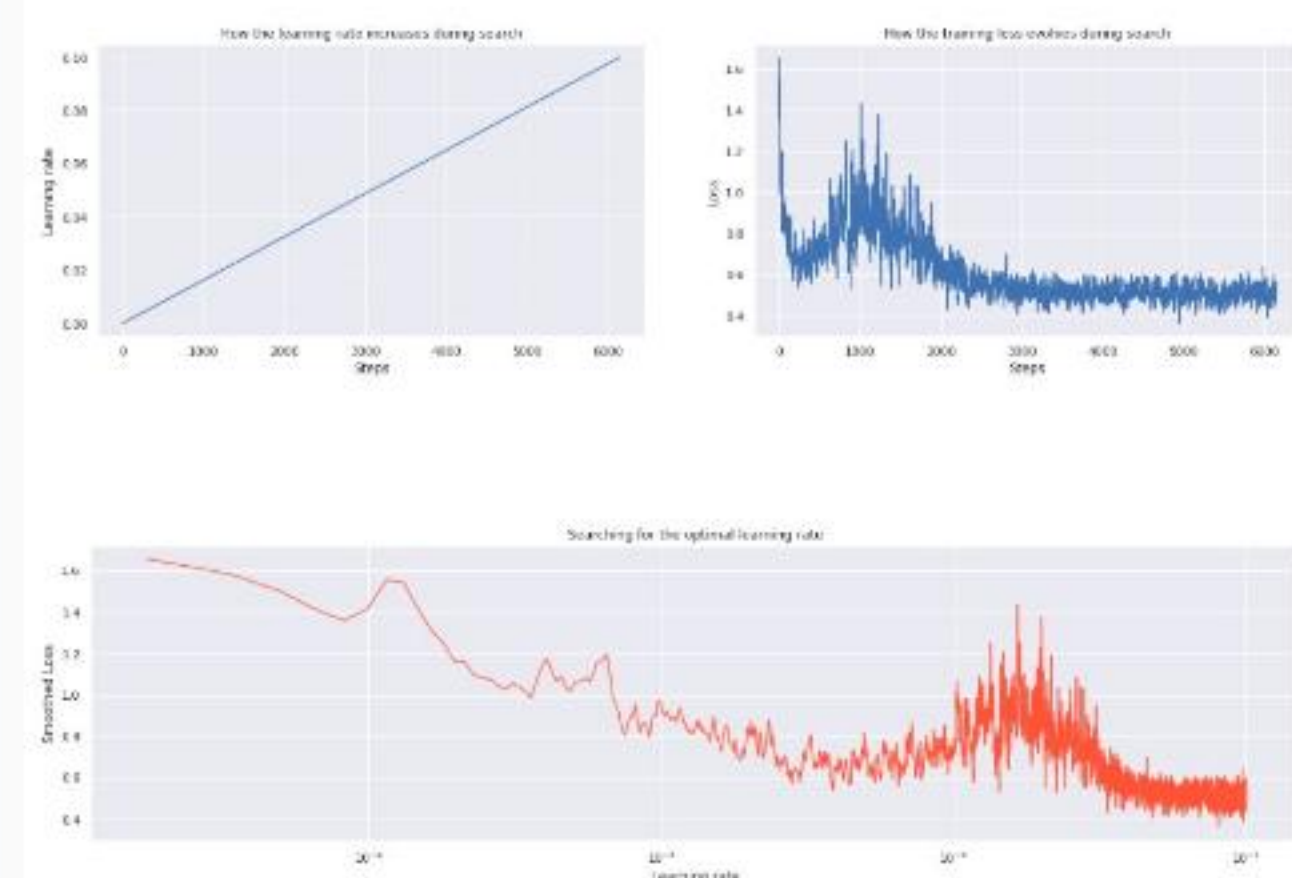
Model Architecture — ResNet-18 CNN



Why ResNet-18?

- **Residual Connections:** Solves vanishing gradients in deep networks.
- **Computational Efficiency:** Lightweight architecture suitable for ~277k patch iterations.
- **Proven Generalization:** Strong baseline performance for biomedical image classification.

Training Strategy & Learning Rate Search

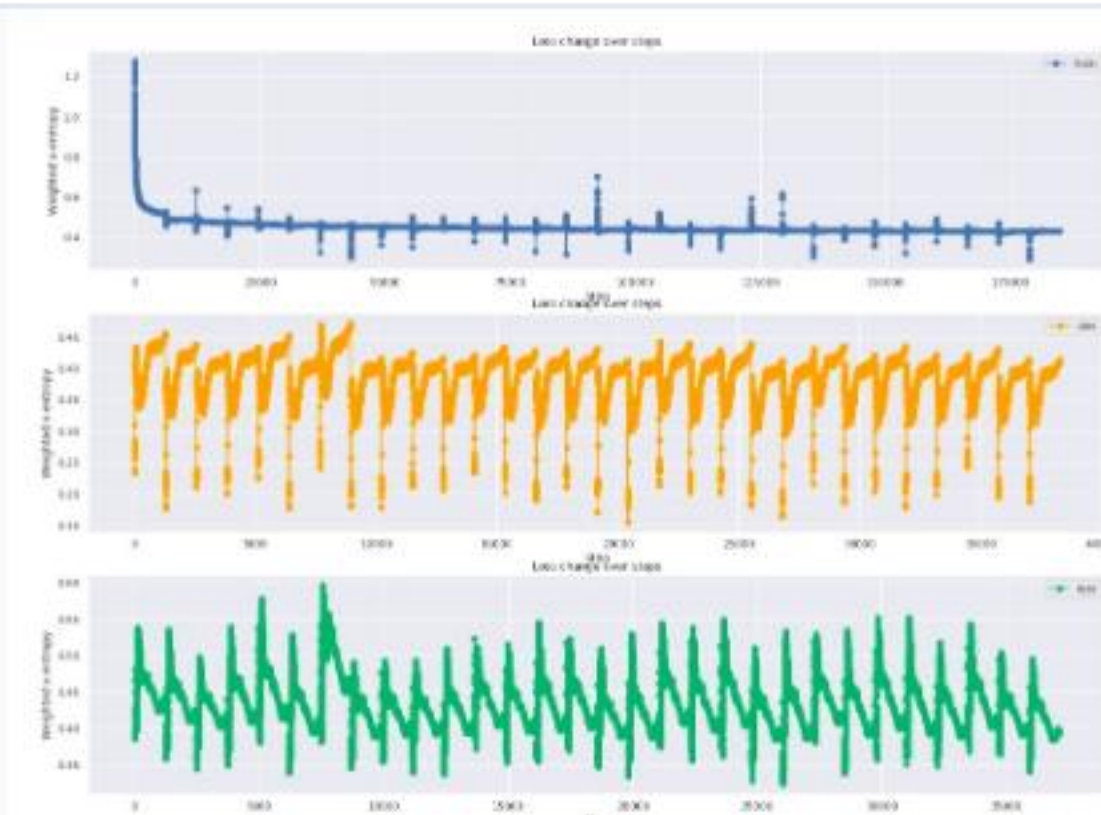


Training Configurations

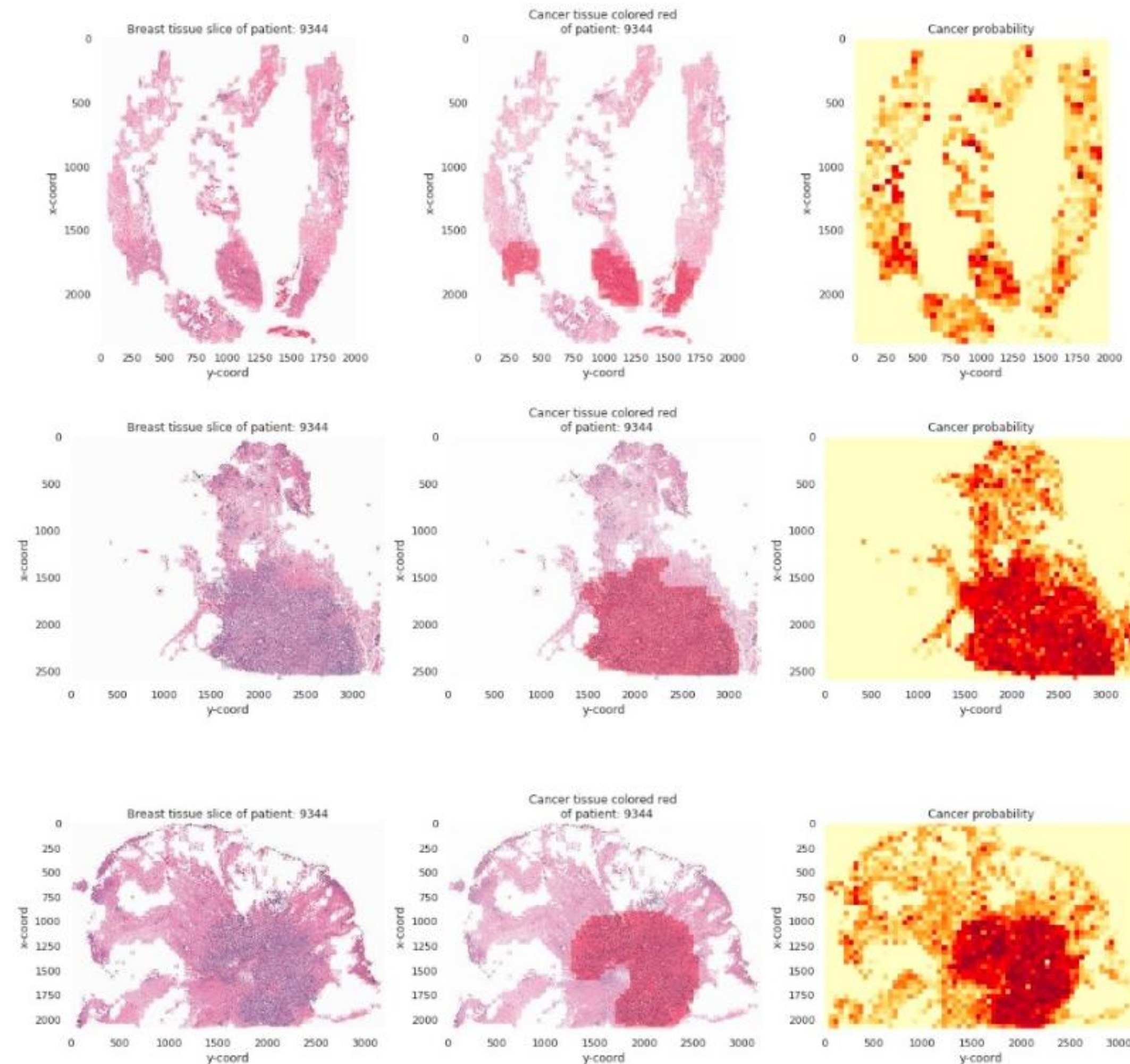
- **Loss:** Class-Weighted Cross-Entropy.
- **Batch Size:** 32.
- **LR Schedule:** Cyclical learning rate strategy.

Evaluation Metrics

Evaluated on **Accuracy, Precision, Recall, and F1-Score** due to dataset imbalance.



Prediction & Tissue-Level Heatmap Reconstruction



Spatial Reconstruction

Patch predictions are re-mapped using spatial (x, y) coordinates to generate continuous cancer probability heatmaps across whole tissue sections.

Pathology Utility

Allows intuitive visualization of localized tumor regions instead of evaluating isolated patches independently.

⚠ Model predictions provide visual screening guidance; they do not replace clinical confirmation.

Results & Key Learnings

What Was Demonstrated

- Built end-to-end histopathology classification pipeline.
- Applied patient-level splitting to prevent data leakage.
- Addressed class imbalance using weighted loss functions.
- Trained ResNet-18 model with cyclical LR strategy.
- Reconstructed spatial tissue heatmaps for visual interpretability.

Technical Competencies

Deep Learning	Computer Vision
PyTorch	Image Processing
Class Imbalance	Model Evaluation

Limitations & Future Work

Current Limitations

- Limited patient sample size (279 patients).
- Stain color variations across slide preparations.
- Absence of external clinical multi-center validation.
- Not certified for independent diagnostic use.

Future Scope

- Incorporate Stain Normalization algorithms (e.g. Macenko).
- Evaluate DenseNet and EfficientNet backbones.
- Implement Grad-CAM for deep feature explainability.
- Test on external multi-site clinical datasets.

✓ END OF PRESENTATION

Thank You

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Questions & Discussion